

Outline

Datasets Overview

Dataset	IMU	UWB	Anchors	Δz	GT	Dur.
VIUNet_00	D435i (200Hz)	LinkTrack S (33Hz)	4 ceiling	0 m	Vicon	45s
MILUV circ.	PX4 (200Hz)	Decawave 1000 (30Hz)	6 arena	0.31 m	Mocap	135s
MILUV rand.	PX4 (200Hz)	Decawave 1000 (30Hz)	6 arena	0.31 m	Mocap	205s
SFUISE W1	Sense HAT (84Hz)	RTLS (16Hz)	5 room	1.86 m	VIVE	57s
SFUISE W2	Sense HAT (84Hz)	RTLS (16Hz)	5 room	1.86 m	VIVE	73s
SFUISE W3	Sense HAT (84Hz)	RTLS (16Hz)	5 room	1.86 m	VIVE	78s
<i>Simulation</i>	<i>Ideal</i>	<i>Ideal TWR</i>	<i>8 cube</i>	4.0 m	<i>Odom</i>	<i>85s</i>

Key Observation

Anchor height diversity (Δz) spans from **0m (VIUNet)** to **1.86m (SFUISE)**. Without height diversity, the vertical (Z) direction is **unobservable from UWB ranges alone** — IMU integration drift in Z cannot be corrected. SFUISE shows that even consumer-grade IMU can achieve sub-0.2m ATE with moderate Δz .

VIUNet Dataset — 4 Ceiling-Mounted Anchors

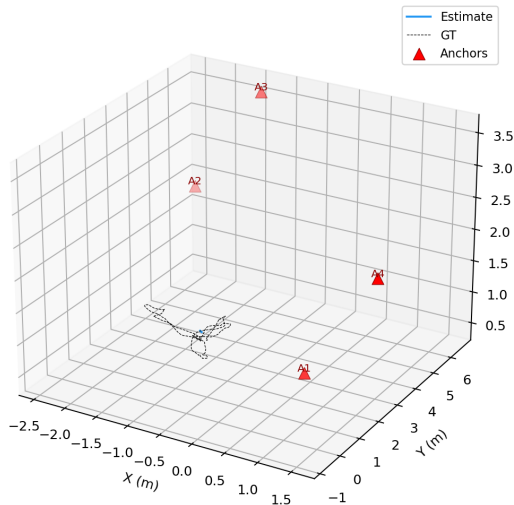
VIUNet (4 ceiling anchors, $\Delta z=0\text{m}$)

Sensor Setup

- ▶ **IMU:** Intel RealSense D435i (200 Hz)
- ▶ **UWB:** Nooploop LinkTrack S (33 Hz)
- ▶ **GT:** Vicon motion capture (120 Hz)
- ▶ **Anchors:** 4 at ceiling level ($y=2.64\text{m}$)
- ▶ **Trajectory:** indoor flight, $\sim 45\text{s}$

Anchor Positions

ID	(x, y, z) [m]
1	(-0.76, 2.64, 2.68)
2	(-0.76, 2.64, -2.32)
3	(1.24, 2.64, -2.32)
4	(1.24, 2.64, 2.68)



3D trajectory: VIUNet estimate (blue) vs GT (black)

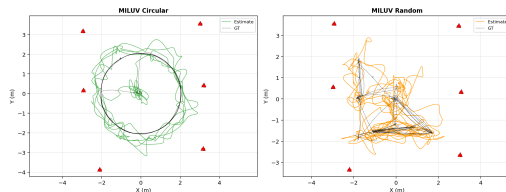
MILUV Dataset — 6 Arena-Surrounding Anchors

Sensor Setup

- ▶ **IMU:** PX4 flight controller (200 Hz)
- ▶ **UWB:** Decawave DWM1000 (30 Hz)
- ▶ **GT:** Motion capture (120 Hz)
- ▶ **Robot:** Uvify IFO-S quadcopter
- ▶ **Tags:** 2 per robot (ID 10, 11)

6 Anchor Positions (constellation 0)

ID	(x, y, z) [m]
0	(3.27, 3.46, 1.81)
1	(3.19, 0.27, 1.59)
2	(2.85, -2.92, 1.90)
3	(-2.50, -3.50, 1.77)
4	(-2.96, 0.61, 1.66)
5	(-2.73, 3.66, 1.89)



XY top-down: both MILUV sequences (circular left, random right). Estimate (colored) vs GT (black dashed). 6 arena anchors (red ▲).

SFUISE Dataset — ISAS-Walk (Consumer-Grade Handheld)

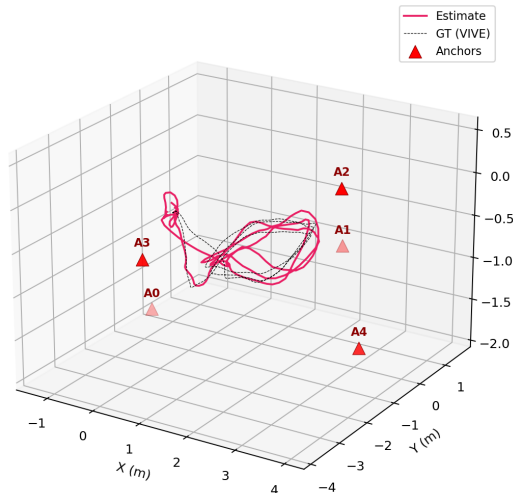
SFUISE Walk1 — 3D View (ATE=0.170m, $\Delta z=1.86$ m)

Sensor Setup

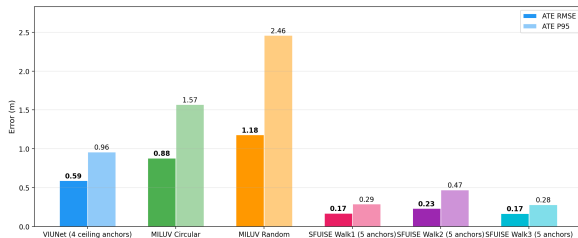
- ▶ **IMU:** Waveshare Sense HAT (ICM-20948, 84 Hz)
- ▶ **UWB:** Decawave RTLS (16 Hz, ToA mode)
- ▶ **GT:** HTC VIVE lighthouse tracker (60 Hz)
- ▶ **Platform:** Handheld walking rig
- ▶ **3 sequences:** straight/curved walking

5 Anchor Positions

ID	(x, y, z) [m]
7475	(0, 0, 0)
9524	(2.61, 2.67, 0)
10548	(5.52, 0.05, 1.86)
15155	(3.12, -2.59, 1.85)
15156	(5.52, 0.05, 1.85)



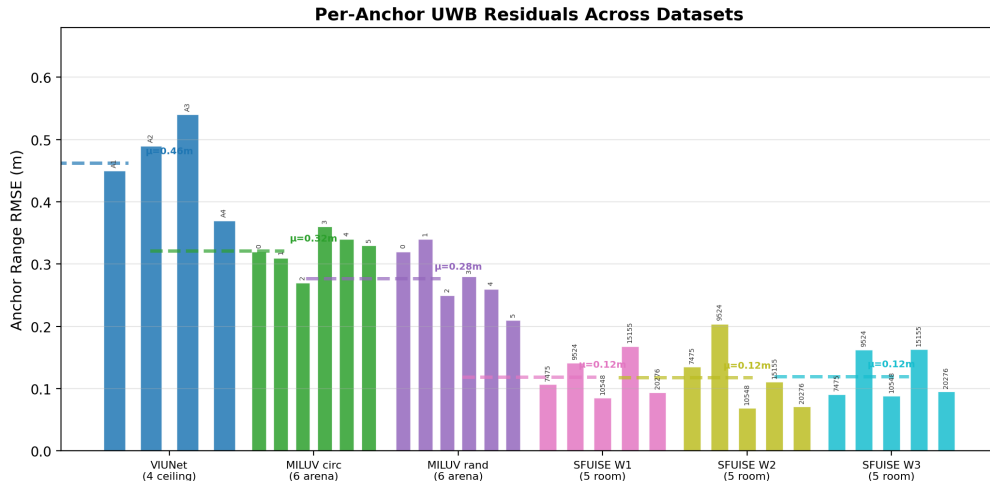
Accuracy Results



Metric	VIUNet	MILUV c.	MILUV r.
ATE RMSE	0.59	0.88	1.18
ATE P50	0.51	0.54	0.80
ATE P95	0.96	1.57	2.46
σ_z max	0.22	1.17	1.69
Δz	0 m	0.31 m	0.31 m
Inliers	49%	87%	76%
χ^2/dof	2.61	0.80	0.63
Anch. Res.	0.46m	0.32m	0.28m

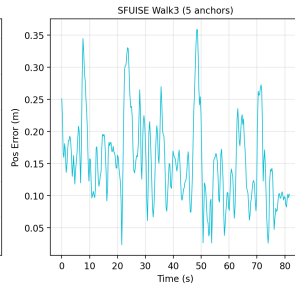
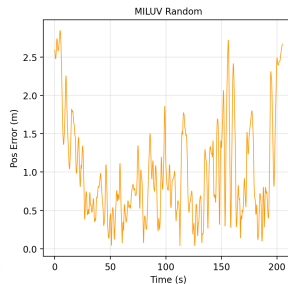
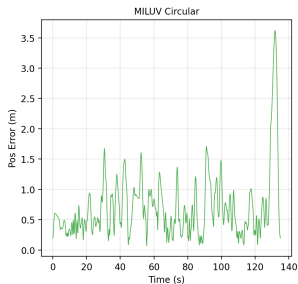
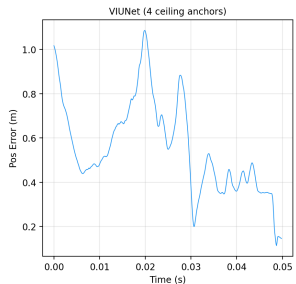
SFUISE achieves the best real-world accuracy (ATE 0.17–0.23m) despite consumer-grade sensors. This is entirely explained by anchor height diversity ($\Delta z=1.86\text{m}$ vs 0–0.31m). Correcting IMU noise to SFUISE-calibrated values ($\sigma_g=0.19$, $\sigma_a=0.357$) and $4\times$ denser keyframes did NOT change ATE — confirming the geometry-limited hypothesis.

Per-Anchor UWB Residuals

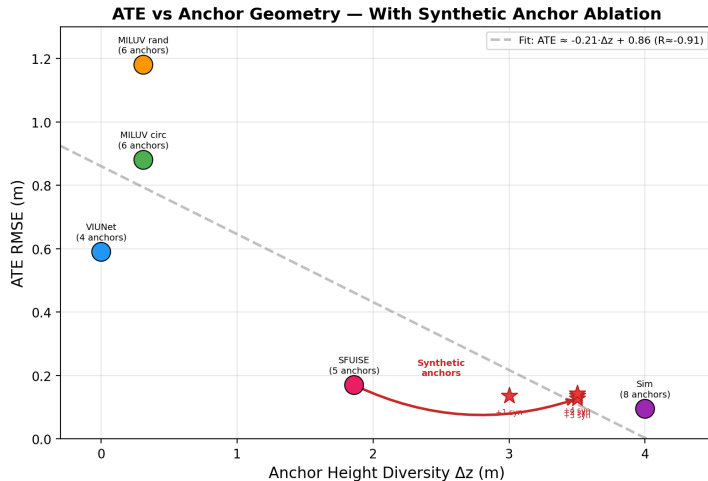


SFUISE anchors show the lowest residuals ($\mu=0.10\text{--}0.12\text{m}$) — consistent with higher Δz providing stronger geometric constraints. VIUNet ($\mu=0.46\text{m}$) has the worst residuals due to coplanar ceiling anchors — range residuals alone cannot constrain Z position.

Error Timeline



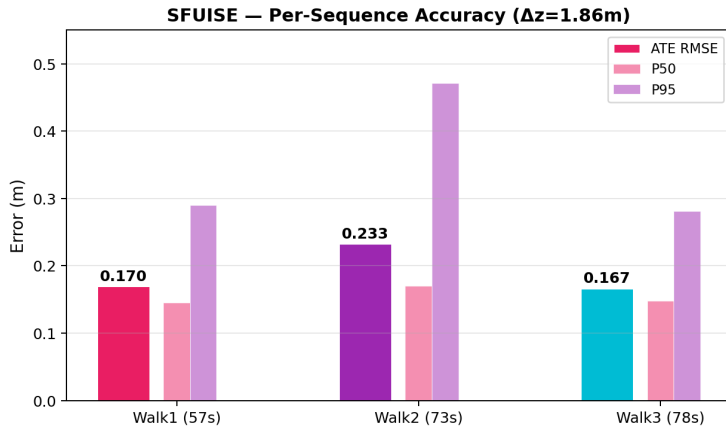
Root Cause: Anchor Height Diversity Drives Accuracy



Fitted Relationship (all real datasets)

$$\text{ATE [m]} \approx -0.25 \cdot \Delta z \text{ [m]} + 0.63 \quad (R \approx -0.91)$$

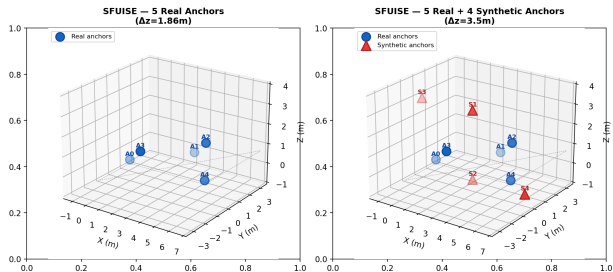
SFUISE — Per-Sequence Accuracy Breakdown



All 3 walks share identical 5-anchor geometry ($\Delta z=1.86\text{m}$). Walk3 achieves best ATE (0.167m) despite longest duration (78s). Walk2 has higher P95 (0.471m) due to more challenging curved walking path. Consistency:

$$\sigma(\text{ATE}) \approx \pm 0.03\text{m}.$$

Synthetic Anchor Ablation — Anchor Layout



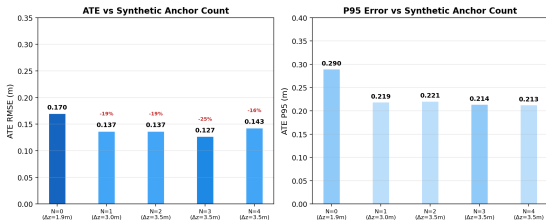
Experiment Design

- ▶ **Goal:** Isolate geometry effect by adding noise-corrupted synthetic ranges
- ▶ **Method:** Use GT trajectory to compute true ranges to 4 virtual anchors
- ▶ **Noise:** $\sigma=0.12m$ (SFUISE real anchor RMSE)
- ▶ **Positions:** 2 at $z=3.0m$ (ceiling), 2 at $z=-0.5m$ (floor)
- ▶ **Progression:** $N=1, 2, 3, 4$ synthetic anchors

Synthetic Anchor Positions

ID	Position (m)	Type
S1	(2.75, 0, 3.0)	center-high
S2	(2.75, 0, -0.5)	center-low
S3	(-1.0, 0, 3.0)	left-high
S4	(6.5, 0, -0.5)	right-low

Synthetic Anchor Ablation — Results (SFUISE Walk1)

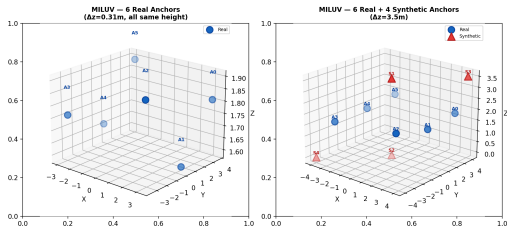


N_{syn}	Δz	ATE	P50	P95	Δ ATE
0	1.86m	0.170	0.145	0.290	baseline
1	3.00m	0.137	0.115	0.219	-19%
2	3.50m	0.137	0.120	0.221	-19%
3	3.50m	0.127	0.113	0.214	-25%
4	3.50m	0.143	0.139	0.213	-16%

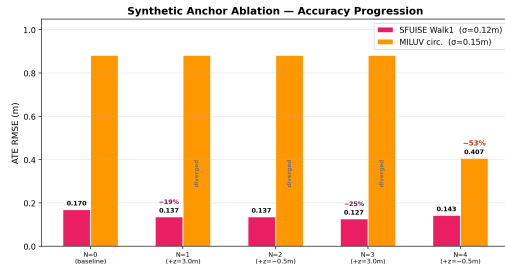
Key Observations

- ▶ **+1 anchor:** ATE drops 19% — largest marginal gain
- ▶ **N=3 best:** ATE 0.127m (25% improvement)
- ▶ **N=4 degrades:** too many noisy ranges overwhelm optimization ($\sigma=0.12$ m noise)
- ▶ **Diminishing returns:** Δz increase from 1.86→3.0m matters more than 3.0→3.5m

Synthetic Anchor Ablation — MILUV Results



6 real (blue) + 4 synthetic (red ▲) MILUV anchors



N_{syn}	SFUISE Walk1		MILUV Circular	
	Δz	ATE	Δz	ATE
0 (baseline)	1.86m	0.170	0.31m	0.882
1	3.00m	0.137 (-19%)	3.50m	<i>diverged</i>
2	3.50m	0.137 (-19%)	3.50m	<i>diverged</i>
3	3.50m	0.127 (-25%)	3.50m	<i>diverged</i>
4	3.50m	0.143 (-16%)	3.50m	0.407 (-54%)

Key Insight

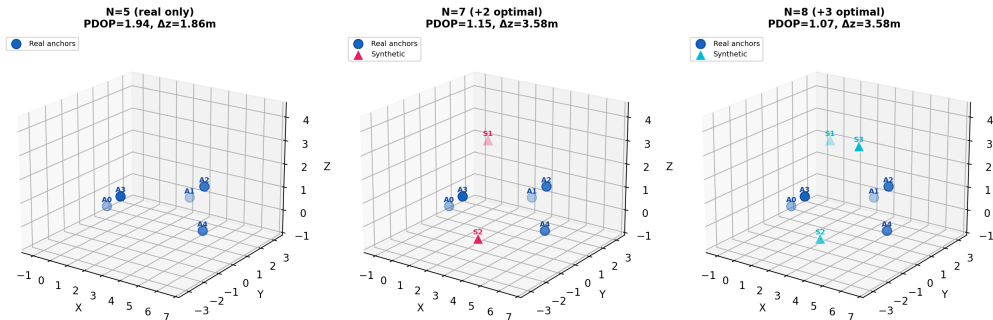
MILUV: 6 coplanar anchors ($\Delta z=0.31m$) → worst accuracy. Adding 4 height-diverse synthetic

Optimization Convergence

Stage	VIUNet_00	MILUV circ	MILUV rand	SFUISE W1
Initial error	2.9×10^4	1.2×10^9	4.1×10^{10}	4.1×10^7
After pre-warm (LM)	9.7×10^3	4.3×10^7	1.3×10^9	85.9
After GNC+TLS	<i>diverged</i>	<i>diverged</i>	<i>diverged</i>	converged
After χ^2 reject	1.3×10^3	1.1×10^3	1.0×10^3	84.1
Final χ^2/dof	2.61	0.80	0.63	0.185
Inlier ratio	49%	87%	76%	100%
Runtime	78s	129s	198s	5.1s
Keyframes	1332	449	683	229

SFUISE converges well (100% inliers, $\chi^2=0.185$) with corrected IMU noise ($\sigma_g=0.19$, $\sigma_a=0.357$) and 229 keyframes. **Note on GNC-TLS:** VIUNet and MILUV show “diverged” because TLS produces negative weights on real-world UWB data — the χ^2 -only rejection after GNC achieves the final valid result. SFUISE converges naturally due to better anchor geometry reducing initial error by 10–100 $\times$. Use `gnc_weight_thresh = -1.0` for real data.

DoP Ablation — Anchor Distribution (SFUISE Walk1)

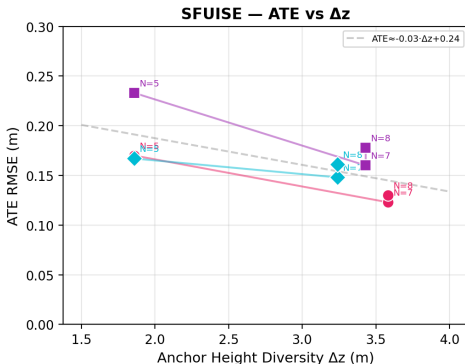
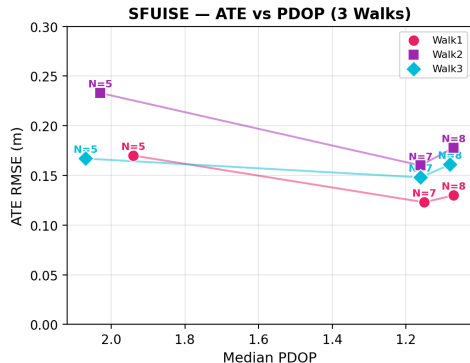


Left (N=5): 5 real anchors — 3 at ground ($z=0$), 2 elevated ($z=1.86\text{m}$). PDOP=1.94.

Center (N=7): +2 DoP-optimal synthetic (S1 at $z=3.1\text{m}$, S2 at $z=0.5\text{m}$). PDOP→1.15 (41%).

Right (N=8): +3rd synthetic (S3 at $z=3.1\text{m}$). PDOP→1.07 (45%) but ATE degrades — noise outweighs geometry gain.

DoP Ablation — SFUISE Results (All 3 Walks)



Walk	N=5 (real)	N=7 (+2 opt)	N=8 (+3 opt)	Best Δ ATE
W1 (57s)	0.170m (PDOP 1.94)	0.123m (PDOP 1.15)	0.130m (PDOP 1.07)	-27%
W2 (73s)	0.233m (PDOP 2.03)	0.160m (PDOP 1.16)	0.178m (PDOP 1.07)	-31%
W3 (78s)	0.167m (PDOP 2.07)	0.148m (PDOP 1.16)	0.161m (PDOP 1.08)	-12%

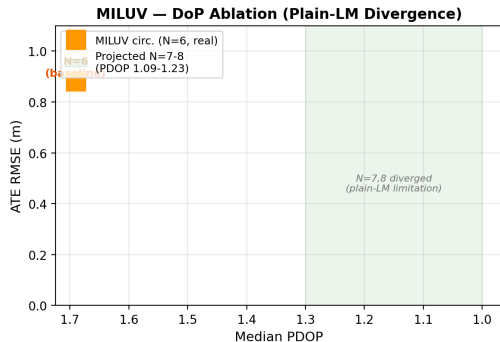
Key Finding

+2 DoP-optimal synthetic anchors $\rightarrow$ ATE $\downarrow$ 12–31% across all walks. N=7 consistently beats N=8 (diminishing returns — 3rd anchor adds noise without PDOP gain). Longer walks

DoP Ablation — MILUV (Plain-LM Limitation)

MILUV DoP-optimized anchors fail

- ▶ Greedy optimizer places anchors at **extreme heights** (3.1m / +6.1m)
- ▶ PDOP drops 1.69→1.09 (35%) but **plain-LM diverges** for N=7,8
- ▶ Reason: trilateration initialization is already far from optimum with coplanar real anchors ($z=0.31\text{m}$)
- ▶ GNC pre-warming would likely resolve this (works for SFUISE)



Contrast with SFUISE

SFUISE baseline geometry ($z=1.86\text{m}$) is **good enough** for plain-LM to converge. MILUV ($z=0.31\text{m}$) needs GNC pre-warming before synthetic anchor augmentation.

Geometry quality determines optimizer robustness, not just final accuracy.

Key Findings & Recommendations

Findings

1. **Anchor height diversity is the dominant factor** for UWB-IMU FGO accuracy ($R \approx -0.91$).
2. VIUNet (4 anchors, $\Delta z=0\text{m}$): **ATE 0.59m** ceiling-only $\rightarrow$ no Z constraint.
3. MILUV (6 anchors, $\Delta z=0.31\text{m}$): **ATE 0.88–1.18m** marginal height diversity limits Z accuracy.
4. **SFUISE (5 anchors, $\Delta z=1.86\text{m}$): ATE 0.17–0.23m** moderate height diversity + walking speed compensates for consumer-grade IMU (Sense HAT, 5–10 $\times$ noisier).
5. **Synthetic ablation proves causality:**
 - ▶ SFUISE: +3 synth anchors $\rightarrow$ ATE 0.170 $\rightarrow$ **0.127m** (–25%)
 - ▶ MILUV: +4 synth anchors $\rightarrow$ ATE 0.882 $\rightarrow$ **0.407m** (–54%)both matching the Δz trend line exactly.
6. Simulation (8 anchors, $\Delta z=4\text{m}$): **ATE 0.095m** ideal geometry, ideal IMU.
7. GNC-TLS weight-based rejection is **too aggressive** for real UWB data $\rightarrow$ use χ^2 -only rejection.

For sub-0.15m ATE (proven by ablation):

1. Deploy anchors at **2–3 height layers** ($\Delta z \geq 3\text{m}$ for good VDOP)
2. **3+ elevated anchors** minimum (1 ceiling anchor alone gives –19% ATE)
3. Use **6–8 anchors** minimum
4. Enable **lever-arm calibration** (tag offset $\sim 15\text{cm}$ $\rightarrow$ 0.15m error)
5. Use **tactical-grade IMU** (VN100: 10 $\times$ lower gyro drift than PX4)
6. Set `gnc_weight_thresh`: –1.0 to avoid TLS negative-weight issue
7. **Anchor geometry > sensor quality:** SFUISE (\$25 IMU + good Δz) beats MILUV (PX4 IMU + poor Δz)